

Mathematics, 2017 EC, entrance pilot test, booklet 070

Grade 12 · Mathematics · 2017 E.C.

Time allowed: 120 minutes

Questions: 38

Total marks: 38

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following fractions correctly expresses $2.343434\dots$? (1)
- A. $\frac{20}{9}$ B. $\frac{23}{9}$
C. $\frac{232}{99}$ D. $\frac{234}{100}$
-
2. Which of the following radical forms is equal to $7^{\left(\frac{2}{3}\right)}$? (1)
- A. $\sqrt{343}$ B. $\sqrt[3]{343}$
C. $\sqrt{49}$ D. $\sqrt[3]{49}$
-
3. What is the standard form of $\frac{1}{50}$? (1)
- A. 0.02×10^2 B. 2.0×10^{-2}
C. 0.02×10 D. 2.0×10^1
-
4. Which one of the following simultaneous equations has a unique solution? (1)
- A. $\begin{cases} y = 2x + 3 \\ -2y = -4x + 6 \end{cases}$ B. $\begin{cases} y = 2x + 3 \\ 2y = 4x + 6 \end{cases}$
C. $\begin{cases} y = 2x + 3 \\ 2y = -4x + 6 \end{cases}$ D. $\begin{cases} y = 2x + 3 \\ -2y = -4x - 6 \end{cases}$
-
5. Which of the following sets contains solutions of $a^2 = a + 2$? (1)
- A. $\{-1, 2\}$ B. $\{-2, 1\}$
C. $\left\{-\frac{1}{\sqrt{2}}, 1\right\}$ D. $\left\{-1, \frac{1}{\sqrt{2}}\right\}$
-
6. How many subsets can be formed from a set with four elements? (1)
- A. 8 B. 15
C. 16 D. 24
-
7. For two sets C and V if $C = \{m, t, h, s\}$ and $V = \{a\}$, what is their symmetric difference? (1)
- A. $\{m, a, t, h, s\}$ B. $\{m, t, h, s\}$
C. $\{\}$ D. $\{a\}$
-
8. Which one of the following statements defines a regular polygon? (1)
- A. It is a concave polygon in which all of its sides are equal and all of its angles are equal.
B. It is a convex polygon in which all of its sides are similar and all of its angles are congruent.
C. It is a convex polygon in which all of its sides are equal and all of its angles are equal.
D. It is a concave polygon in which all of its sides are similar and all of its angles are congruent.
-
9. The degree measurement of an angle of $\frac{5}{6}\pi$ radian equals to: (1)
- A. 120 B. 130
C. 160 D. 150

10. Which of the following is a symmetrical property of a circle? (1)
- A. The circle is symmetrical about any of its diameters.
 - B. The circle is symmetrical about any of its radii.
 - C. The circle is symmetrical about any of its chords.
 - D. The circle is symmetrical about any of its arcs.
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11. A circular bread of radius 20cm is chopped into eight equal slices. What is the area of one slice of bread? (1)
- A. $2.5\pi \text{ cm}^2$
 - B. $5\pi \text{ cm}^2$
 - C. $45\pi \text{ cm}^2$
 - D. $50\pi \text{ cm}^2$
-
12. Which of the following is NOT a purpose of statistical mode? (1)
- A. To identify the top customer in a shop.
 - B. To identify the highest mark in an examination.
 - C. To recognize the frequent visiting tourists in a park.
 - D. To recognize the redundancy of car accidents in a city.
-
13. If a die is rolled twice, then what is the probability of the sum of the numbers appearing is 5? (1)
- A. $\frac{1}{36}$
 - B. $\frac{1}{18}$
 - C. $\frac{5}{36}$
 - D. $\frac{1}{9}$
-
14. Given a triangle with coordinates (2, 0), (0, 3), and (4, 5). Which of the following is a position vector of a side of the triangle? (1)
- A. $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$
 - B. $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$
 - C. $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$
 - D. $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$
-
15. Which one of the following intervals contains a zero of $f(x) = 1 - x^2$? (1)
- A. $(-3, \frac{-3}{2})$
 - B. $(\frac{1}{2}, 2)$
 - C. $(\frac{-1}{2}, \frac{1}{2})$
 - D. $(2, 3)$
-
16. Given a function $f(x) = 8x^3 - 27$. Which one is a zero of $f(x)$? (1)
- A. $\frac{3}{2}$
 - B. $\frac{1}{5}$
 - C. $\frac{2}{3}$
 - D. $\frac{2}{7}$
-
17. Let a and b are non-zero real numbers. Let m and n are integers. Then, which one is the correct law of exponents? (1)
- A. $a^m \times a^n = a^{m \times n}$
 - B. $\frac{a^m}{a^n} = a^{\left(\frac{m}{n}\right)}$
 - C. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
 - D. $(ab)^m = a^m + b^m$
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18. The simplified form of $\log_9 \sqrt[3]{54} - \log_9 \sqrt[3]{2}$ equals to: (1)
- A. $-\frac{1}{3}$
 - B. $\frac{1}{3}$
 - C. $-\frac{1}{2}$
 - D. $\frac{1}{2}$
-
19. Which one is true about graphs of $f(x) = \left(\frac{2}{5}\right)^x$ and $g(x) = \left(\frac{5}{3}\right)^x$? (1)
- A. They have the same x-intercept.
 - B. They have the same y-intercept.
 - C. They are decreasing over $(-\infty, 0)$.
 - D. They are increasing over $(0, \infty)$.

20. What is the solution set for the equation $\left(\frac{125}{27}\right)^{x+1} = \left(\frac{9}{25}\right)^{3x-1}$? (1)
- A. $\{-\frac{1}{9}\}$ B. $\{\frac{5}{3}\}$
C. $\{-2\}$ D. $\{2\}$
21. Which one of the following is the longest side of a quadrilateral with vertices $P(-3, -3)$, $Q(3, -1)$, $R(1, 2)$ and $S(-5, 1)$? (1)
- A. PQ B. QR
C. RS D. SP
22. If a point R divides $\overline{PQ}$ internally where $P(1, -3)$ and $Q(2, 5)$ in 1:2 ratio, what are the coordinates of R? (1)
- A. $(\frac{5}{3}, \frac{7}{3})$ B. $(\frac{4}{3}, \frac{11}{3})$
C. $(\frac{4}{3}, -\frac{1}{3})$ D. $(2, \frac{14}{3})$
23. Which of the following pairs of lines is perpendicular to one another? (1)
- A. $x - y = 33$; $x - y = 44$ B. $y = 345$; $y = -345$
C. $2x - y = 0$; $x + 3y = 0$ D. $x = 567$; $y = -567$
24. Which one of the following trigonometric values is correct? (1)
- A. $\tan(330^\circ) = \frac{-1}{\sqrt{3}}$ B. $\tan(150^\circ) = \frac{1}{\sqrt{3}}$
C. $\tan(210^\circ) = -\sqrt{3}$ D. $\tan(30^\circ) = \sqrt{3}$
25. Which one of the following is true about trigonometric values? (1)
- A. $\sin(-150^\circ) = \frac{1}{2}$ B. $\sin(-210^\circ) = \frac{1}{2}$
C. $\cos(-150^\circ) = \frac{\sqrt{3}}{2}$ D. $\cos(-210^\circ) = \frac{\sqrt{3}}{2}$
26. If θ is an angle in standard position and $P(x, y)$ is a point on the terminal side of θ , what is the cosecant value of θ ? (1)
- A. $\csc \theta = \frac{\sqrt{x^2+y^2}}{y}$ B. $\csc \theta = \frac{\sqrt{x^2+y^2}}{x}$
C. $\csc \theta = \frac{y}{\sqrt{x^2+y^2}}$ D. $\csc \theta = \frac{x}{\sqrt{x^2+y^2}}$
27. If two chords intersect at the center of a circle and create four congruent arcs, what is the degree measure of each angle formed by the chords? (1)
- A. 45° B. 90°
C. 180° D. 360°
28. What is the volume of a ball with diameter 30 cm? (1)
- A. $4,500\pi \text{ cm}^3$ B. $36,000\pi \text{ cm}^3$
C. $9,000\pi \text{ cm}^3$ D. $900\pi \text{ cm}^3$
29. If a regular frustum of pyramid with 6 m by 6 m and 2 m by 2 m square bases has slant height of $2\sqrt{65}$ m, what is its lateral surface area? (1)
- A. $72\sqrt{65} \text{ m}^2$ B. $64\sqrt{65} \text{ m}^2$
C. $12\sqrt{65} \text{ m}^2$ D. $32\sqrt{65} \text{ m}^2$

30. A frustum of right circular cone is 4 cm high and has bases of radii 5 cm and 3 cm, what is its volume? (1)

- A. $\frac{136}{3}\pi \text{ cm}^3$ B. $\frac{196}{3}\pi \text{ cm}^3$
C. $\frac{100}{3}\pi \text{ cm}^3$ D. $\frac{86}{3}\pi \text{ cm}^3$

31. Which of the following sentences defines a power function? (1)

- A. $f(x) = a \cdot x^r$ where a and r are constant real numbers and $a \neq 0$.
B. $f(x) = a \cdot x^r$ where a and x are constant real numbers and $r \neq 0$.
C. $f(x) = a \cdot x^r$ where x and r are constant real numbers and $r > 0$.
D. $f(x) = a \cdot x^r$ where a, x and r are constant real numbers and $x > 0$.

32. If a signum function is given by $f(x) = 3 - 3\text{sgn } x$, then which of the following is true? (1)

- A. The domain of f is $\{-1, 0, 1\}$. B. The domain of f is $\{0, 3, 6\}$.
C. The range of f is $\{-1, 0, 1\}$. D. The range of f is $\{0, 3, 6\}$.

33. Which of the following is true about the graph of $f(x) = \frac{x+2}{x-2}$? (1)

- A. $y = 1$ is the horizontal asymptote. B. $y = -2$ is the horizontal asymptote.
C. $x = 1$ is the vertical asymptote. D. $x = -2$ is the vertical asymptote.

34. Below is the graph of a rational function f of x . (1)

If it does not cross $y = x$, which of the following is true?

[Figure: A Cartesian coordinate grid showing the horizontal 'x - axis' and vertical 'y - axis'. The line $y = x$ passes through the origin into quadrants I and III. A continuous curve labeled 'graph of function f ' lies entirely above the line $y = x$, opening upwards in a U-shape. Its minimum is on the positive y-axis, and as x increases into the first quadrant, it rises alongside and above the line $y = x$.]

- A. If $x \rightarrow -\infty, f(x) \rightarrow y = x$ B. If $x \rightarrow +\infty, f(x) \rightarrow y = x$
C. $f(-300) < -300$ D. $f(300) < 300$

35. What type of matrix is $\begin{pmatrix} 3 & 0 \\ 0 & -3 \end{pmatrix}$? (1)

- A. Identity matrix B. Scalar matrix
C. Skew-symmetric matrix D. Symmetric matrix

36. Given points $A(2, 3)$, $B(1, -2)$ and $C(0, 4)$, what is the value of $\vec{AB} + 2\vec{BC} - \vec{AC}$? (1)

- A. $\begin{pmatrix} -4 \\ 2 \end{pmatrix}$ B. $\begin{pmatrix} 4 \\ -5 \end{pmatrix}$
C. $\begin{pmatrix} -1 \\ 6 \end{pmatrix}$ D. $\begin{pmatrix} 1 \\ -6 \end{pmatrix}$

37. Which of the following is a correct property on vector dot products? (1)

- A. $3 \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right] = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot 3 \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ B. $3 \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right] = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} -1 \\ 0 \end{pmatrix}$
C. $\begin{pmatrix} 0 \\ -1 \\ -1 \\ 0 \end{pmatrix} \cdot \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right] = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} +$ D. $\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right] \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} +$
 $\left[\begin{pmatrix} -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right]$

38. If a triangle of vertices $P(1, 2)$, $Q(3, 4)$ and $R(0, 2)$ is rotated about the origin by 30° counterclockwise, the vertices of the image triangle are: (1)

A. $P' = \left(\frac{\sqrt{3}-2}{2}, \frac{1+2\sqrt{3}}{2} \right); Q' = \left(\frac{3\sqrt{3}-4}{2}, \frac{3+4\sqrt{3}}{2} \right); R' = (-1, \sqrt{3})$

C. $P' = \left(\frac{\sqrt{3}-2}{2}, \frac{1+\sqrt{3}}{2} \right); Q' = \left(\frac{\sqrt{3}-1}{2}, 1 + \frac{\sqrt{3}}{2} \right); R' = \left(\frac{3\sqrt{3}}{2} - 2, \frac{3+2\sqrt{3}}{2} \right)$

B. $P' = \left(\frac{\sqrt{3}-2}{2}, 1 + \sqrt{3} \right); Q' = \left(\frac{\sqrt{3}-2}{2}, 1 + \sqrt{3} \right); R' = (1, -\sqrt{3})$

D. $P' = \left(\frac{\sqrt{3}-2}{2}, \frac{1+\sqrt{3}}{2} \right); Q' = \left(\frac{\sqrt{3}-2}{2}, 1 + \sqrt{3} \right); R' = \left(\frac{3\sqrt{3}-4}{2}, \frac{3+2\sqrt{3}}{2} \right)$

— END OF EXAMINATION —